
Replay Accuracy Is Not Enough: An Audited Closed-Loop Evaluation Protocol for Learned Robot State Observers

Anonymous Author(s)

Affiliation

Address

email

Abstract

Robot observers are usually selected by estimation error on logged trajectories, although their outputs are consumed by feedback controllers. We audit this model-selection practice for dead reckoning, an extended Kalman filter (EKF), and GRU- or selective-state-space residual observers. Six principal observers are evaluated on paired replay and closed-loop protocols across 24 conditions spanning sensing blackouts, range noise, landmark sparsity, and gyro bias. Replay position RMSE preserves broad ordering (Spearman $\rho = 0.923$) but selects a suboptimal closed-loop observer in 5/24 conditions. We then test three controller-aware offline alternatives rather than assuming that task awareness solves the problem. A local controller-sensitivity metric reduces the observed failures to 3/24, but its bootstrap interval overlaps no improvement and its worst selection regret is larger; exact command disagreement and counterfactual error replay perform worse. Matched EKF-anchor ablations overturn a confounded original conclusion: selectivity is regime-dependent, while the analytic anchor is the more consistent stabilizer. Three training initializations further reveal large variability under long blackouts. Finally, we evaluate 600 seconds of UTIAS MRCLAM physical logs and a real-noise-calibrated closed-loop replay, exposing the poor map transfer of the ordered landmark-slot representation. The resulting contribution is an open evaluation protocol: report replay and closed-loop metrics, top-1 agreement, regret, matched ablations, training-seed uncertainty, and physical-log stress tests.

1 Introduction

A state estimate is rarely the final output of a robot. It is fed to a controller, which changes the future state and observations. Yet localization and learned-filter systems are commonly selected by open-loop (OL) position or heading error on fixed logs. This assumes that lower replay error implies a better closed-loop (CL) system. The assumption is convenient, but it ignores controller sensitivity, temporal error structure, and the endogenous distribution shift created by feedback. Closely related proxy failures have been studied in model-based reinforcement learning (Lambert et al., 2020); here we isolate the problem at the state-observer layer.

We consider a differential-drive robot tracking a figure-eight from biased odometry and intermittent range-bearing measurements. Our candidate set spans classical and learned designs: dead reckoning, an EKF (Thrun et al., 2005), and recurrent residual observers with GRU or compact selective diagonal SSM cores inspired by S4 and Mamba (Gu et al., 2022; Gu and Dao, 2024). The recurrent correction is anchored either to dead reckoning or the EKF, following the broader hybrid-filter principle of retaining algorithmic structure while learning difficult components (Becker et al., 2019; Jonschkowski et al., 2018; Revach et al., 2022).

36 This work began as an architecture paper. A code and claim audit changed the contribution. The
 37 original top-1 mismatch statistic omitted GRU-EKF, and its mask/selectivity ablations changed the
 38 analytic anchor at the same time. We correct both problems and ask a broader question: *what evidence*
 39 *is needed to select a learned observer for feedback deployment?* Our contributions are:

- 40 • a paired replay/closed-loop benchmark with six deployment candidates, 24 degradation
 41 conditions, and top-1 selection regret;
- 42 • an empirical comparison of pose RMSE with exact command disagreement, local control
 43 sensitivity, and counterfactual error replay;
- 44 • matched EKF-anchor ablations and a three-training-seed robustness study;
- 45 • physical-log localization and zero-shot transfer on UTIAS MRCLAM (Leung et al., 2011),
 46 plus a closed-loop replay calibrated from its sensor residuals and observation timing.

47 The result is deliberately not a claim that one offline scalar solves deployment selection. Instead, it
 48 identifies which conclusions survive an audit and releases an evaluation checklist that makes failure
 49 visible.

50 2 Setup and evaluation protocol

51 **Robot, sensing, and controller.** The state is $s_t = (x_t, y_t, \theta_t)$ with unicycle dynamics integrated
 52 at 0.1 s. A pure-pursuit controller (Coulter, 1992) acts on the estimated pose. Odometry includes
 53 multiplicative slip, angular-rate bias, and Gaussian noise. Exteroception consists of known-map
 54 range-bearing observations, with an active landmark subset and periodic complete blackouts.

55 **Observers.** The six main candidates are DR, EKF, GRU-DR, SSM-DR, GRU-EKF, and SSM-
 56 EKF. Learned observers predict a small recurrent pose residual on an analytic anchor. Inputs contain
 57 odometry increments, ordered masked landmark tuples, an anchor encoding, and the EKF covariance
 58 diagonal where applicable. The SSM is a compact input-gated diagonal recurrence, not a full Mamba
 59 block. Models are trained on 200 domain-randomized trajectories of 200 steps.

60 **Paired protocols.** In OL replay, every observer receives identical odometry and measurements
 61 generated along an expert-controlled trajectory. In CL evaluation, the controller consumes the
 62 candidate estimate, so observer error changes actions and future inputs. Four physical axes—blackout
 63 duration, range noise, active landmark count, and gyro bias—are swept over six levels with ten
 64 evaluation seeds. The primary OL and CL metrics are position RMSE and cross-track RMSE.

65 For condition c , proxy q , and observer set $\mathcal{O}$, the selected observer and regret are

$$o_{q,c}^* = \arg \min_{o \in \mathcal{O}} q_c(o), \quad \mathcal{R}_{q,c} = J_{\text{CL},c}(o_{q,c}^*) - \min_{o \in \mathcal{O}} J_{\text{CL},c}(o). \quad (1)$$

66 This separates broad correlation from the actual deployment decision.

67 **Controller-aware proxies.** Let $e_t = \hat{s}_t \ominus s_t$ and $\pi(s, t)$ be a time-indexed controller. Exact
 68 command disagreement measures $\|D^{-1}(\pi(\hat{s}_t, t) - \pi(s_t, t))\|$. Our local control sensitivity error
 69 (LCSE) is

$$q_{\text{LCSE}} = \left[\frac{1}{T} \sum_t \|D^{-1} J_\pi(s_t, t) e_t\|_2^2 \right]^{1/2}, \quad (2)$$

70 where J_π is a finite-difference controller Jacobian and D normalizes action units. We also test
 71 counterfactual error replay: a shadow plant is propagated by the controller after injecting the
 72 candidate’s logged error trace. All are computed without rerunning sensing or the observer in closed
 73 loop.

74 3 Results: no log-only scalar replaces closing the loop

75 Table 1 confirms a nuanced result. Position RMSE is a strong coarse proxy, but it changes the
 76 top-1 observer in 20.8% of conditions. LCSE improves the observed winner count, suggesting that

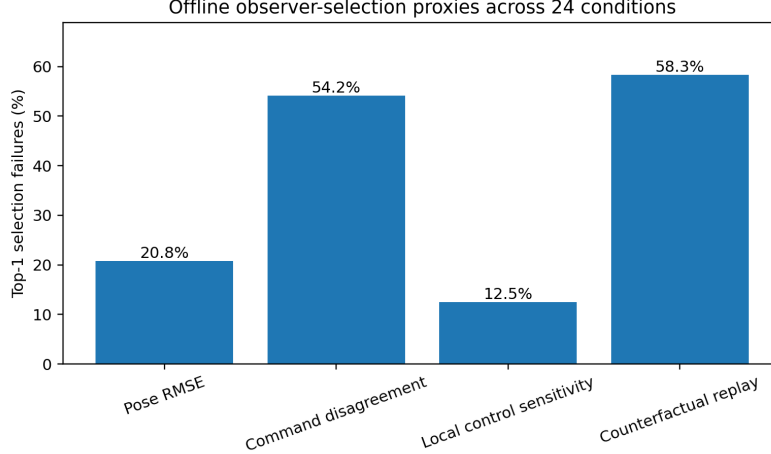


Figure 1: Observed top-1 selection failures over 24 conditions. LCSE reduces the count from 5 to 3, but a condition bootstrap gives a wide interval for the reduction; exact command disagreement and counterfactual replay are worse.

Table 1: Offline proxies against closed-loop cross-track RMSE. Regret is in meters.

Proxy	Spearman ρ	Failures	Mean regret	Max regret
Position RMSE	.923	5/24	.0051	.0347
Command disagreement	.876	13/24	.0188	.1208
LCSE	.903	3/24	.0051	.0940
Counterfactual replay	.842	14/24	.0159	.0474

77 controller anisotropy contains useful information. However, its worst regret is driven by the two-
78 landmark condition, and a paired condition bootstrap gives a 95% interval of $[-0.125, 0.292]$ for the
79 reduction in failure fraction. It would therefore be overclaiming to call LCSE a solved replacement.
80 More surprisingly, exact command disagreement performs poorly: local action closeness does not
81 capture how estimation errors modify future state and sensing. Counterfactual error replay also fails
82 because a fixed logged error trace does not adapt to the shadow trajectory. These negative baselines
83 are useful: adding a controller to a metric does not automatically make it deployment-valid.

84 The corrected six-observer result also retains a sharp within-model counterexample from the original
85 study. Under compound stress, SSM-EKF reduces OL position RMSE by 43% relative to the EKF
86 ($0.829 \rightarrow 0.472$ m) while increasing CL cross-track error by 10% ($1.324 \rightarrow 1.459$ m). GRU-EKF is
87 best in both columns, so the defensible statement is not that SSMs are inferior, but that a large replay
88 improvement for one observer need not transfer to control.

89 4 What actually stabilizes the learned observer?

90 The original ablation compared a selective EKF-anchored model with non-selective and mask-free
91 models anchored to dead reckoning. That design confounded recurrence with the strength of the
92 analytic prior. Figure 2 retrains both ablations on the same EKF anchor. Nominal cross-track RMSE
93 is 0.199 m for selective+mask, 0.203 m without selectivity, and 0.195 m without the mask feature.
94 Under long blackouts, the non-selective model is better (0.235 vs. 0.336 m; paired bootstrap CI for
95 the difference $[-0.182, -0.020]$ m), while under high gyro bias it is worse (0.393 vs. 0.314 m; CI
96 $[0.011, 0.145]$ m). Thus selectivity is a regime-dependent bias-robustness tradeoff, not a universal
97 stability requirement. The consistent qualitative factor is the physics anchor: DR-anchored learned
98 observers remain much less reliable in the loop.

99 **Training-seed uncertainty.** Three SSM-EKF initializations give similar nominal means (0.189–
100 0.199 m), but long-blackout means range from 0.231 to 0.336 m and high-noise means from 0.249

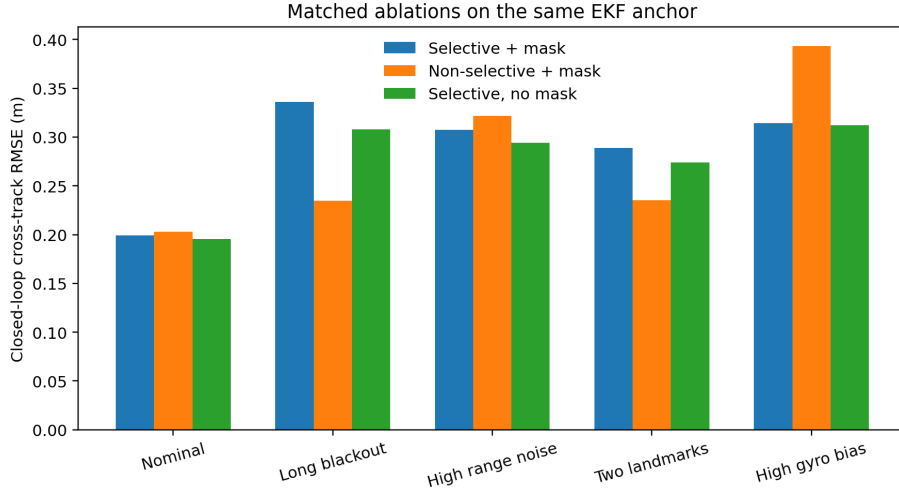


Figure 2: Matched EKF-anchor ablations. Selectivity is beneficial under high gyro bias, but non-selectivity is better under long blackouts and two-landmark sensing. Mask removal is usually close to the full model.

101 to 0.307 m. A single checkpoint therefore supports nominal claims more strongly than stress-regime
 102 claims. Evaluation-seed confidence intervals alone do not capture this uncertainty.

103 **Controller sensitivity.** In a matched five-seed secondary study, replacing pure pursuit with a
 104 Kanayama controller changes the closed-loop oracle observer in 8/24 conditions. Replay position
 105 RMSE misselects the Kanayama observer in 7/24 conditions with maximum regret 0.092 m; a paired
 106 seed bootstrap gives a wide $[0.208, 0.625]$ interval for the oracle-change fraction. Thus observer
 107 selection is partly controller-specific, although this lower-seed analysis is a sensitivity rather than a
 108 primary result.

109 5 Physical-log and real-noise stress tests

110 UTIAS MRCLAM provides timestamped odometry, range-bearing observations, known landmark
 111 positions, and Vicon ground truth (Leung et al., 2011). We align 600 s of Dataset 1 Robot 1 to 6,000
 112 steps, containing 1,878 landmark observations. A simple 15-landmark EKF covariance sweep reaches
 113 1.357 m position RMSE, compared with 3.392 m for dead reckoning (Figure 3).

114 The original learned representation has exactly eight ordered landmark slots and was trained on one
 115 synthetic map. We therefore perform a deliberately difficult zero-shot test on an eight-landmark
 116 MRCLAM subset. SSM-EKF and GRU-EKF achieve 2.625 m and 2.597 m position RMSE, modestly
 117 better than dead reckoning on that subset (2.822 m), but their heading RMSE is worse (1.65–1.66 rad
 118 vs. 1.51 rad). This is not evidence of successful real-world transfer. It exposes the next architecture
 119 requirement: permutation-invariant innovation-set encoding and randomized-map training.

120 Finally, we estimate empirical odometry and measurement residuals and preserve the physical
 121 measurement-count sequence in a simulated closed-loop replay. In this low-bias log, dead reckoning
 122 is the closed-loop oracle among an EKF covariance grid, while controller-aware proxies choose
 123 aggressive EKFs and incur 0.12–0.20 m regret. The result reinforces the main message: even
 124 a physically calibrated proxy can mis-rank observers when it does not model the full feedback
 125 adaptation.

126 6 Discussion and evaluation checklist

127 The useful field contribution is not a new scalar score but a reproducible protocol:

- 128 1. evaluate every deployment candidate in both common-input replay and closed loop;

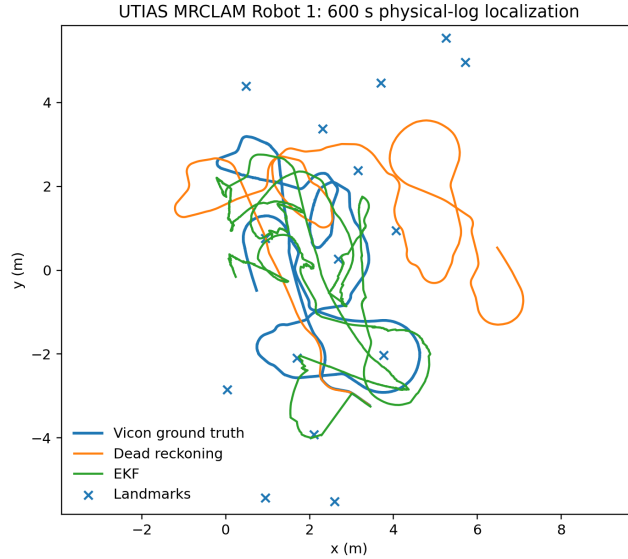


Figure 3: Physical-log localization on UTIAS MRCLAM Robot 1. This is real sensing and Vicon ground truth, not a closed-loop hardware experiment.

2. report rank correlation, top-1 agreement, and absolute/relative selection regret;
3. use matched anchors, inputs, and capacity in ablations;
4. separate environment-seed and training-seed uncertainty;
5. test task-aware offline proxies as baselines, including their failure cases;
6. include physical logs and clearly distinguish real sensing, calibrated simulation, and hardware control;
7. audit whether the representation can vary map geometry, cardinality, and landmark order.

The current benchmark remains limited to known data association, a known map, only two controller families, and simulated closed-loop plant dynamics. More controller seeds and gain shifts, multiple MRCLAM robots/datasets, randomized-map set encoders, and hardware trials are the highest-value next steps.

7 Conclusion

Replay position RMSE is informative but incomplete for selecting learned state observers. It preserves broad rankings while changing the deployment winner in 5/24 conditions. Controller-aware offline metrics expose useful sensitivity structure but do not consistently replace rollouts. Matched ablations show that the analytic anchor is more reliably stabilizing than selectivity, which trades off robustness across regimes. Training-seed and MRCLAM experiments further reveal uncertainty and map-transfer limitations hidden by the original formulation. The practical recommendation is to treat observer evaluation as a decision problem and close at least a small, stratified loop before deployment.

References

- S. Thrun, W. Burgard, and D. Fox. *Probabilistic Robotics*. MIT Press, 2005.
- R. C. Coulter. Implementation of the pure pursuit path tracking algorithm. Technical Report CMU-RI-TR-92-01, Carnegie Mellon University, 1992.
- P. Becker, H. Pandya, G. Gebhardt, C. Zhao, C. J. Taylor, and G. Neumann. Recurrent Kalman networks. In *ICML*, 2019.
- R. Jonschkowski, D. Rastogi, and O. Brock. Differentiable particle filters. In *RSS*, 2018.

- 155 G. Revach et al. KalmanNet: Neural network aided Kalman filtering for partially known dynamics. *IEEE*
156 *Transactions on Signal Processing*, 70:1532–1547, 2022.
- 157 A. Gu, K. Goel, and C. Ré. Efficiently modeling long sequences with structured state spaces. In *ICLR*, 2022.
- 158 A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *COLM*, 2024.
- 159 N. Lambert, B. Amos, O. Yadan, and R. Calandra. Objective mismatch in model-based reinforcement learning.
160 In *L4DC*, pages 761–770, 2020.
- 161 K. Y. K. Leung, Y. Halpern, T. D. Barfoot, and H. H. T. Liu. The UTIAS multi-robot cooperative localization
162 and mapping dataset. *The International Journal of Robotics Research*, 30(8):969–974, 2011.